

Python Major Project – Initial Proposal Form (Agile SDLC)

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3D-DASH

A remastering of hit 2013 game 'Geometry Dash' in a 3D medium

Project Summary

3D-Dash is a remaster of the 2013 side-scrolling platformer known as 'Geometry Dash' created by Robert 'RobTop' Topala. The original game features a square shaped player traversing through a series of levels littered with various obstacles in order to reach an end goal and switch to the next level.

Similar to the design of the game's inspiration, 3D-Dash will feature a cuboid individual that progresses linearly through a series of levels through the act of platforming by various means. However, it's variance comes from it's application of the 3D medium rather than the original game's 2D environment. While both games feature the player being locked into a constant forward momentum in order to maintain the pace and music of the game, this recreation will apply the 3rd dimension to the player character, thus requiring individuals to consider the game on a deeper level and with more forethought to an action at any given moment.

With various mechanics to leverage the environment, including a five-lane switching system, jumping, gravity changes and other means of traversal, players will need to think critically about situations as they arise in order to successfully traverse the environment without colliding with obstacles such as spikes and walls that would otherwise forcefully restart the level.

Problem Statement

In the gaming environment of today, many programs struggle to define a fast-paced environment while also incorporating aspects of quick-action problem solving and reaction time. While 3D-Dash by no means can be seen as a learning experience, it's focus, and the focus of similar games, (ie Geometry Dash), consider the player's enjoyment of a fast-paced environment, balanced with the difficulty of navigating a situation in an unconventional or previously unconsidered way.

Challenging the player to think 'outside the box' in a shortened time window can lead to frustration in the moment, but ultimately allows for further freedom of self-expression, especially in the 3d medium where players can discover multiple routes to a solution that were significantly less obvious. Thus, 3D-Dash allows for self-expression through on-the-fly problem solving, tapping into a medium that is underused in the gaming market.

Target Audience/Users

3D-Dash naturally taps into the market of people previously interested in the Geometry Dash title and it's various subprojects (ie Geometry Dash World, SubZero, etc), as part of it's connection to the platformer genre. Despite my own lack of skill in 3D design as a result of this being my first project, the three dimensions provided by my model of the game allow for a higher level of fast paced action while also giving room for those with less skill in reaction time to enjoy the game.

Stemming from my level design philosophy of scaling difficulty over the course of both a level and the progression of the game as a whole, different portions of the game will thus cater to different skill levels, simultaneously decreasing the barrier for entry while raising the ceiling for skill expression in the game. As a result, the target audience will involve all skill levels of those interested in the platformer genre,

Proposed Features

Functional Proposed Components

- Five-lanes of interconnected routes to progress, creating a fully 3D environment that allows for quick thinking and problem solving to move past complex obstacles
- Manoeuvrable camera that can be used to orient and view different angles, as well as provide new perspectives as the player traverses.
- Multiple varied levels with different mechanics and traversal systems to ensure interesting and flexible gameplay
- Autosaved high-score system that tracks your best attempts in each level
- Vibrant effects that change the way each level looks with pulsating lighting changes and graphics

Nonfunctional Proposed Components

- Invigorating custom-designed (CITATION NEEDED) music that matches up with in-game movements and provides an extra layer of interest to the design of the levels.

- Player character customization systems, such as adjusting the object of use, changing colours, or even different textures and patterns.
- Unlockable game features for finding secrets within levels, allowing for further replayability in the game.

Technologies / Tools

The proposed systems for this project are subject to change based on desired implementation of features, but the current systems for this project are:

- Python 3.10x – 3.12x
- Ursina 3D Rendering Layer (boilerplate systems for Panda3D)
 - o Panda 3D Rendering Engine
- Blender 3D Graphics Design
- Various audio editing and compositing tools including:
 - o Audacity
 - o MuseScore Studio
 - o MuseHub
 - o Bandlab

Method / Approach

My work within this project will be structured similarly to a fully structured Agile workflow, complete with proper testing and feedback from ‘hired’ gameplay testers, as well as simulated launches to ensure the game is in a proper state for submission and completion.

Requirements and Analysis will be the first to be conducted, including this project proposal as part of my own research. In this project, the main requirements will be specified by my own research into the target market, including critical reviews of platformer genres to find commonly desired components and pre-existing similar products from which I can observe common tropes and design elements to utilise. I will also be conducting an analysis of my own capabilities and that of the engine I have chosen to develop with, in their respective ability to achieve certain imagined goals.

Designing the project itself will be especially documented, including references to my inspiration and potential avenues for further development should time allow for it. Designing the project will happen multiple times throughout the project but the majority of my own considerations will be developed at the beginning, preceding any proper development of the systems. Later design activities will be more focused on individual features based on user feedback, as well as my own availability and workload overall.

Implementation of designed features will be conducted first with attempts at various components of the engine I assigned, in order to determine the capabilities I can address within the project itself, without relying too heavily on needing a finished product. Later development progress will be through the collective of systems I have designed, and in a significantly more

structured format that utilizes important OOP principles, especially that of Inheritance and Polymorphism in order to optimise and improve the functionality of the game itself.

Testing and implementation/development components will be repeatedly occurring over the process of 'sprints', being individually set ranges of the timeline in which that is the primary focus of the development of the project. Testing systems will involve not only the isolation and confirmation of different critical components, but also testing the entire program as a whole and confirming my findings with short Q/A tests by 'hired' helpers who have experience in game design or gameplay systems.

Maintenance and mock launches will also be included as repeating cycles at the end of development sprints and during testing phases through tests of packaging and release systems (e.g. how can the system handle conversion to an .exe file for release), as well as further functionality tests from both myself and any in-person tests I can find. I especially insist upon finding other people to help test, as despite my own qualifications for gameplay testing, different people will have different ideas for different situations, and my own experience in developing the game has the potential to cloud my own ability to notice flaws.

Major Anticipated Challenges

The primary expected difficulties of this project are expected to arise in three specific areas, as detailed below. My understanding of these points was defined by a series of rapid tests and early programming sprints designed as part of my feasibility study into the capabilities of the software systems I was planning on testing.

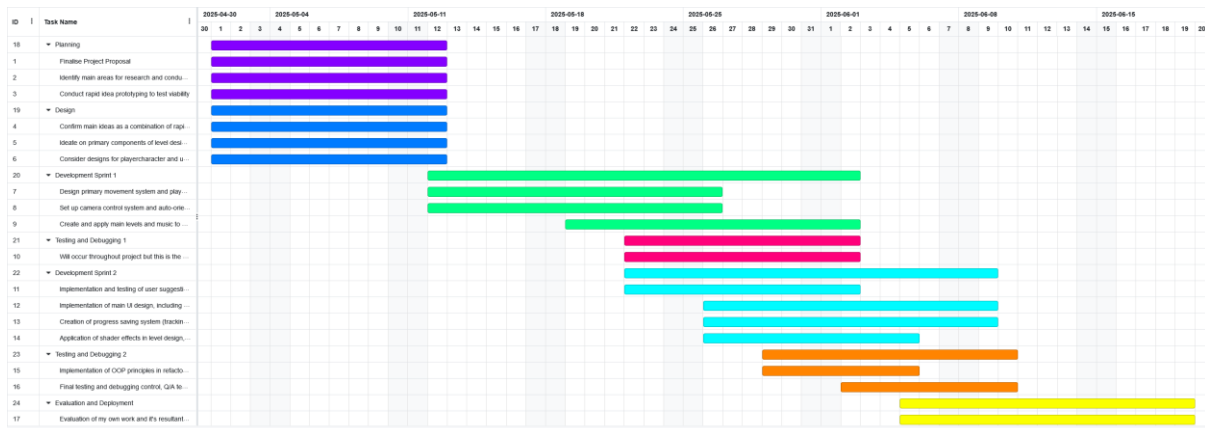
Firstly, the progression of my knowledge within the 3D medium thus requires me to understand the Ursina systems on a deep level in order to execute my level of customisation to the program. From basic testing since starting the writing process of this project proposal, it has already been found that I must design my own movement and collision for the player character, rather than using a preset system, as a result of my more complicated movement and camera mechanisms. Sorting the finer details of this system, especially in collision interactions between the models I plan on importing, will be a significant difficulty, and managing the creation of my own system of mesh interactions for complex models will be a difficult process.

The second expected challenge for this project lies in the music system. My plans for the creation and integration of music are heavily based on the original Geometry dash's systems, where music is synchronized to certain actions and jumps within each of the levels. In this area, it becomes rather difficult to have my own version of this, as I cannot simply create an in-game framework due to the logistical limitations of my own time, the 3d medium, and the Ursina engine. Thus, I need to utilise the modelling software I was considering (blender), and systematically consider where the player would be at certain points in the song and link it to the progression of the level during construction.

The third challenge I will no doubt find within this project is the adoption of a variety of new mediums outside of my normal areas. In previous projects, I was primarily focused on 2D games or assistive softwares, so I have extremely limited experience within the 3D medium, aside from my previous experience with Blender software and modelling. Furthermore, I have almost zero experience in making music using computer-based technology, and will thereby need to teach myself the systems in order to design my own music to work with the levels (as I would prefer to minimise free/stolen asset use where possible).

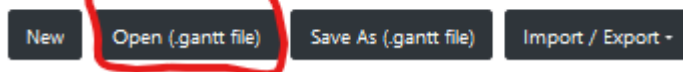
Within all of these issues remains the core problem of learning and understanding the intricacies behind the newer mediums I am pursuing. Resolving these issues and maintaining my personal mental state through these challenges will require, above all, proper time management and personal restraint to avoid constantly following new ideas and getting wrapped up in unreasonably intricate detailing. Rapid prototyping and testing will be extremely helpful as well, enabling me to quickly explore multiple ideas and understand their feasibility over the longer time period of the term.

Project Timeline Overview



In case this is too difficult to read, I have attached a link to a google drive file. It contains the full Gantt chart image, two excel files (one with the full gantt chart table and one with a summarised version) and a file with the .gantt extension that can be imported into a site like [S LINK](#) in order to be interpreted into the full gantt chart with dropdown sequencing and full name explanations.

* This 100% free tool allows you to save your gantt charts on your computer as .gantt f



site)

(Where to import the .gantt folder on the

[DRIVE FOLDER LINK](#) (Automatically set to share to anyone with the link)